PKG-Net: Physical Knowledge Feature-Guided Learning for Aircraft Detection in Optical Remote Sensing Images

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***Abstract*— Existing aircraft detection methods primarily rely on loss function constraints to guide the learning of end-to-end detectors, which significantly diverges from the judgment logic of human experts. Inspired by how human experts make decisions based on cognitive features, this article introduces the concept of physical knowledge features. Leveraging three properties of physical knowledge features, we identify the circle grayscale (CG) feature of aircraft and propose a physical knowledge-guided network (PKG-Net). By embedding CG features into the supervised learning process, the network improves its proficiency in learning stable features, thereby enhancing detection accuracy. Within this network, a multiscale circular frequency filter module (MS-CFFM) is responsible for extracting and integrating aircraft CG features across different scales. Adaptive channel selection module (ACSM) selectively activates channels for learning CG features. The hybrid attention feature fusion module (HAFFM) focuses intensively on the central localization of aircraft and deep reinforcement of channels. Experimental results on the RSOD and UCAS-AOD datasets demonstrate that the proposed method surpasses existing techniques in accuracy, achieving state-of-the-art performance.**

***Index Terms*— Aircraft detection, physical knowledge feature, remote sensing (RS).**

I. Introduction

As TECHNOLOGY advances, the methods for remote sensing (RS) image analysis have become more sophisticated and advanced [1], [2], [3]. Object detection in RS images has been extensively applied in various fields such as surface observation [4], [5] and urban planning [6]. Among many observed objects, aircraft serve as crucial transportation means in civilian domains and as key strike forces in the military.

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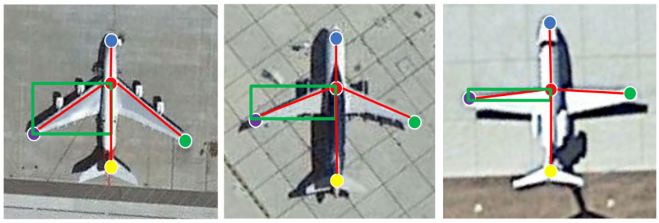


Fig. 1. Example of a cognitive feature in human interpretation. The rectangularity of the shapes formed by the starting and ending points of wings varies across different aircraft models.

However, due to the diversity in aircraft sizes, variability in attitudes, and the complexity of backgrounds, aircraft detection remains a challenging research topic [7].

Current mainstream aircraft detection methods focus on building and training an end-to-end detector [8]. They enhance detection robustness under varying environmental conditions by adding invariant constraints on image textures, colors, and so on to the trained loss function. Although these mainstream methods yield satisfactory results under certain conditions, their feature extraction relies on the distribution and statistical patterns of historical data, thus necessitating a large number of samples for training.

We argue the issue with the current end-to-end detectors stems from a significant divergence between their construction logic and the judgment habits of human experts [9]. Observations show that humans do not strictly rely on basic image features such as texture and color for object location and classification. These features can easily become confused under complex environmental changes. For example, the same type of aircraft can exhibit significant differences in texture and color under different lighting and atmospheric conditions. Conversely, human experts rely more on stable object features that align with cognitive characteristics. For example, the topological relationships of key points vary among different aircraft models, and the rectangularity between key points of the wings and fuselage also differs, as shown in Fig. 1. However, it is challenging for current detectors to learn these types of features through end-to-end training. In this article, we define such features as physical knowledge features. Through systematic research, we have summarized that physical knowledge features useful for object detection and recognition should have the following three properties.

1) *Uniqueness:* This reflects the unique attributes that distinguish a category of objects from others, or the unique

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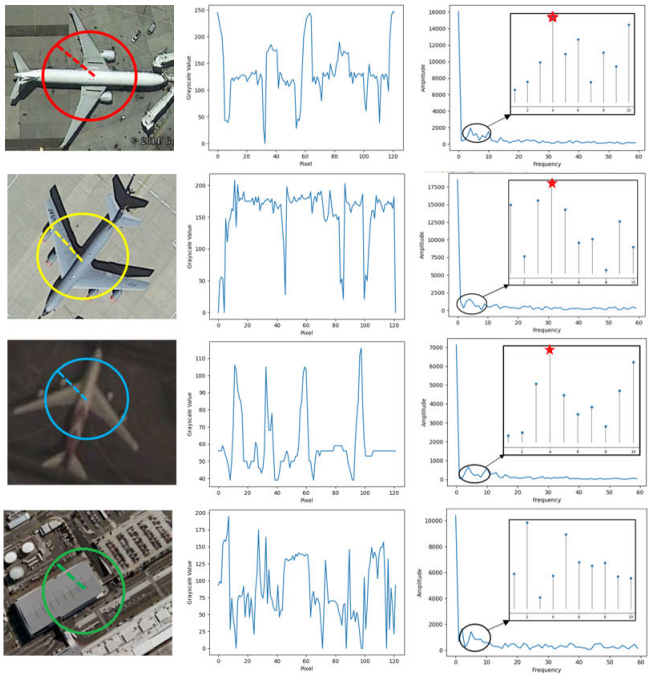


Fig. 2. Under conditions of good lighting, gray paint, and heavy fog, the circular sampling sequence of the aircraft reaches its maximum at a Fourier series of 4. This feature is not observed in randomly selected complex backgrounds.

properties that can effectively differentiate between different models within the same category. Learning unique features enables the model to quickly identify the boundary conditions that distinguish an object category from others or recognize minor yet crucial differences when making fine-grained distinctions within the same class.

2) *Quantifiability:* These features can be modeled and quantitatively expressed through mathematical or physical formulas. Quantifiable characteristics allow the model to adjust parameters effectively through optimization algorithms like gradient descent, ensuring accurate capture of key features.
3) *Stability:* These features remain stable across varying sample sizes, background environments, posture angles, or time conditions (stability refers to the feature values staying constant, maintaining within a specific range, or exhibiting regular changes). Stable features reduce the model's overfitting to a specific dataset, lowering the risk of overfitting.

It needs to be emphasized that based on the above characteristics, researchers can be guided to discover more qualified physical knowledge features.

Based on the analysis above, this article finds a feature for optical RS images of aircraft that meets the aforementioned criteria—circle grayscale (CG) feature [10].

1) *The CG feature is a unique characteristic of aircraft:* Aircraft have fixed geometric shapes with four main protrusions: the head, tail, and two wings. As shown in Fig. 2, with the aircraft's center as the circle's center and choosing a circumference larger than the width of the fuselage but smaller than the wingspan, the pixel grayscale values around this circumference will change regularly from dark to light, repeating four times. The background does not have this feature.
2) *The CG feature can be quantifiably represented using Fourier series:* Since the grayscale values of the pixel sequence sampled along the circumference centered on the aircraft exhibit a trend of four peaks and four valleys, this implies the presence of a frequency component that is one-quarter of a period in this sequence. From the mathematical perspective of the discrete Fourier transform (DFT), at a frequency index of 4, the DFT shows the highest amplitude response, as illustrated in Fig. 2.
3) *The CG feature exhibits significant stability:* According to DFT properties, when the starting position of the sampled sequence of circumference pixels changes, it is equivalent to a phase shift in the sampling sequence, which does not affect the Fourier transform's amplitude response. Therefore, under different background environments and different rotation angles of the aircraft, as shown in Fig. 2, the maximum amplitude response value can be obtained at a frequency index of 4, indicating that the CG feature has excellent invariance to rotation, texture, lighting, and other factors.

The three properties of the CG feature have been verified above, but the existing algorithms for extracting CG features from images need to process each pixel in the image one by one, resulting in a very high computational cost. Also, the aircraft in the RS image may be of different sizes, and the existing algorithms may not be able to efficiently capture the features of all sizes of the aircraft. If multiple scales of the circle frequency filter (CFF) are designed, the computational cost will increase proportionally with the increase in scale. Therefore, designing an algorithm to quickly and efficiently compute the CG feature of aircraft in images with multiple scales also needs to be considered.

Building on the analysis above, to enable models to learn and extract such physical knowledge features, this article introduces a network guided by a physical knowledge network (PKG-Net). As shown in Fig. 3, unlike general detectors, the proposed network processes the input image through two routes: a backbone network for extracting general features and a parallel path that captures specific physical characteristics of aircraft, steering the backbone network's learning. Moreover, the network architecture integrates an advanced feature fusion module designed to amplify the interactivity of the channels and spatial dimensions of the feature maps. (Our goal is to provide researchers with a general method by which networks can learn stable features of the object under the guidance of physical knowledge, thereby enhancing the model's ability to understand the object. Therefore, our approach is not limited to dual-branch networks.)

In PKG-Net, to tackle the high cost of pixel-by-pixel computation and the limitations of a single scale, PKG-Net introduces the multiscale circular frequency filter module

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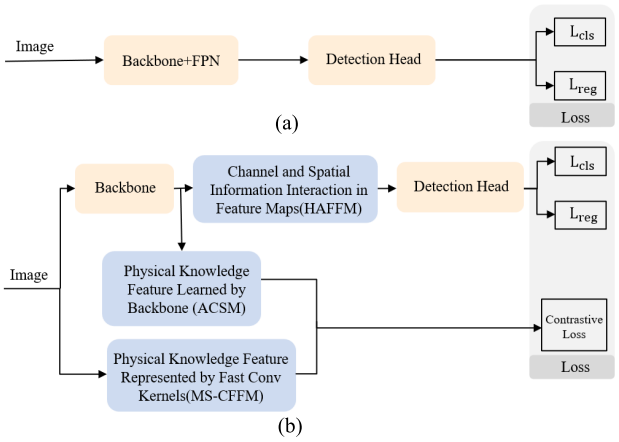


Fig. 3. Comparison between the general detector workflow and the proposed PKG-Net workflow. (a) General detector and (b) our physical knowledge-guided detector.

(MS-CFFM), which uses convolutional kernels of various scales to effectively extract CG feature and swiftly generate feature maps. To address the issue of Swin Transformer's overemphasis on CG features, we incorporate the adaptive channel selection module (ACSM), activating specific channel features at key stages in the network to balance the learning of CG features and other semantic features. In addition, we propose the hybrid attention feature fusion module (HAFFM) to enhance the model's ability to locate and refine boundaries against complex backgrounds. Through quantitative and qualitative experiments, as well as comparisons with other detectors, the effectiveness of the proposed methods has been validated. PKG-Net significantly enhances aircraft detection performance, reducing the gap between limited-sample and ample-sample scenarios.

The contributions of this article are summarized as follows.

1) Our systematic research and analysis have led to the definition of a stable feature set for object detection, categorized into three properties: uniqueness, quantifiability, and stability. This framework can guide researchers in exploring object features based on physical knowledge and their application in directing network learning. To our knowledge, this marks the first international proposal of the concept of physical knowledge-guided learning.
2) To facilitate the learning of the aforementioned object features in existing models, we propose PKG-Net, a network designed to learn under the guidance of physical knowledge. This network assists the Swin Transformer in understanding and extracting the physical knowledge features of aircraft.
3) To address the high time consumption and single-scale issue in traditional CG feature extraction, we design the MS-CFFM. Furthermore, to reduce redundancy from adjusting all channels during CG feature learning, we propose ACSM, allowing the network to autonomously select which channels to activate.
4) To ensure that the feature maps effectively express the knowledge provided by the CG feature, we design the HAFFM. This module enhances the ability of feature maps to capture essential information, such as object locations.

II. Related Work

In recent decades, aircraft detection technology has rapidly evolved. This section will briefly review some of the work related to object detection and aircraft detection.

A. Object Detection Methods

Object detection methods based on convolutional neural networks (CNNs) are primarily categorized into two types: region-based two-stage detectors and regression-based one-stage detectors [11]. Two-stage detectors, such as RCNN [12], Fast RCNN [13], and Faster RCNN [14], first identify potential candidate regions within an image and then extract and classify features from each region. Conversely, one-stage detectors, including the YOLO series [15], [16], [17], [18], and SSD [19], omit the candidate region selection step, directly applying regression analysis and classification predictions on prediction boxes.

In recent years, the introduction of Transformer models [20], [21], [22] into the domain of object detection has offered a fresh perspective. Specifically, the Swin Transformer [23], [24], which adapts the Transformer structure for visual tasks, employs a hierarchical Transformer to capture image features across different scales. This advancement significantly enhances object detection performance by efficiently processing multiscale visual information [25], [26], [27].

The majority of these object detection methods apply to aircraft detection. However, these end-to-end networks, which learn the mapping from input to output directly through the constraints of a loss function, have limitations and challenges. They often fail to fully integrate domain knowledge and primarily rely on the distribution of training data, leading to a strong dependency on data. The networks depend on extensive annotated data for training, which is manually labeled by human experts. Inspired by these limitations, we propose a novel approach: leveraging the reasoning logic of human experts (i.e., the physical knowledge feature defined in Section I) to guide network learning.

B. CG Feature

In exploring the physical knowledge feature of aircraft, researchers have identified a unique pattern in the circumferential pixel values around the center of aircraft, known as the CG feature. Cai and Su [28] introduced that at an appropriate radius, these circumferential pixel values present a special pattern. This pattern, characterized by a gray value curve with a sequence of 4–5 peaks and 4–5 troughs, is referred to as the CG feature.

Building on the CG feature, some researchers design the CFF algorithm to extract candidate regions of aircraft [29]. After CFF was proposed, many researchers used CFF to extract candidate regions of aircraft and perform secondary screening on the candidate regions. An et al. [30] used the

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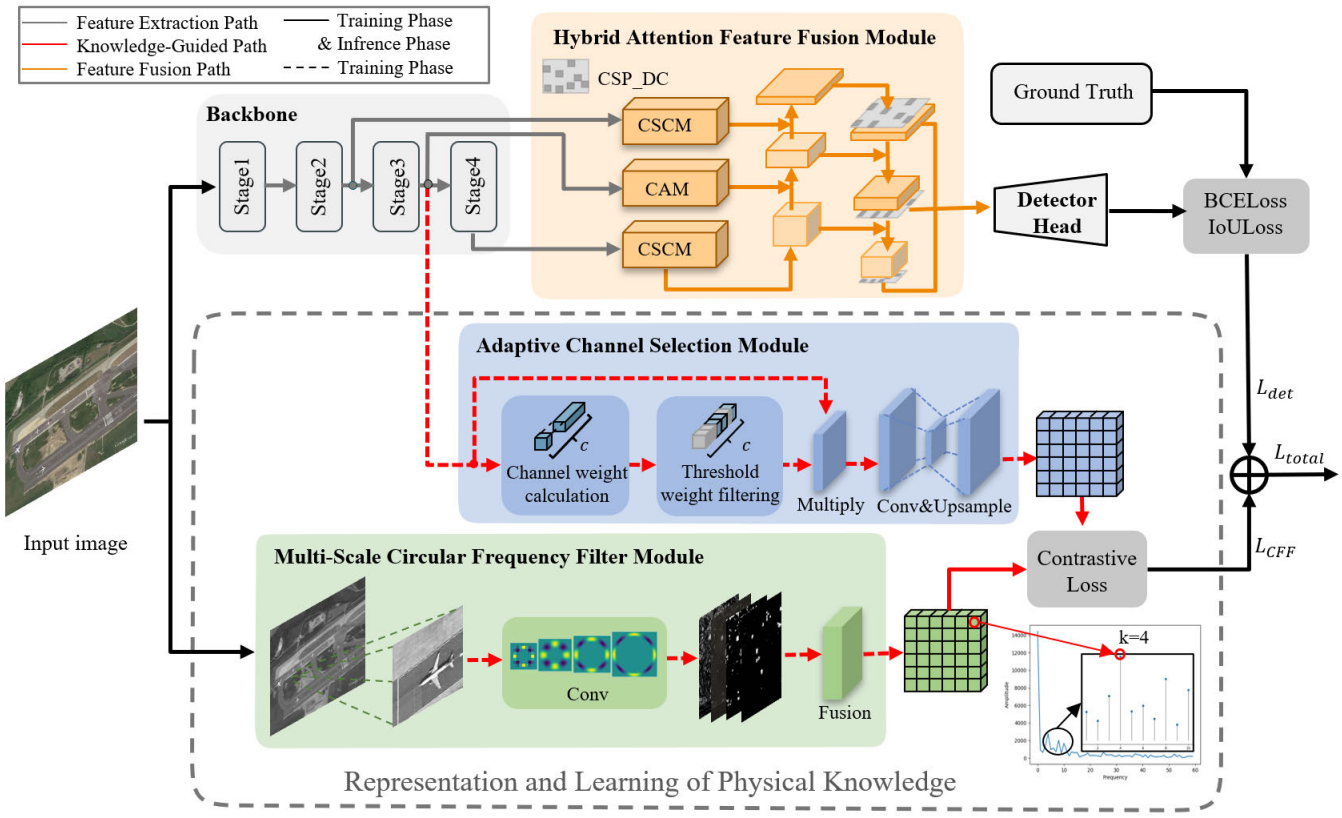


Fig. 4. Architecture of PKG-Net. The input image is processed through two distinct branches. The gray branch (Feature Extraction Path) utilizes the Swin Transformer for feature extraction. The red branch (Knowledge-Guided Path) specializes in extracting multiscale CG features of the image through MS-CFFM and then supervises the output of the third stage of the first branch using ACSM. In the orange branch (Feature Fusion Path), features extracted by the backbone network are fused and enhanced through HAFFM. Finally, the network performs aircraft classification and localization through its head. The solid lines in the figure represent the paths used during training and inference, and the dashed lines represent the paths used during the training phase only.

CFF algorithm to extract candidate regions of aircraft and then input the histogram of the oriented gradient of these candidate regions into the AdaBoost classifier for aircraft detection. Liu and Shi [31] used the CFF algorithm to extract the candidate regions, used radial gradient transform and sparse coding to combine new features, and then input the new features into a linear support vector machine for candidate region screening. However, these methods do not adequately address the problem of aircraft scale variations and can be both time-consuming and labor-intensive when applied to large datasets.

To overcome these limitations, Wang et al. [10] proposed an end-to-end lightweight aircraft detection framework called CGC-NET. This method utilizes the circumferential gray values of different scales as vectors as input, extracts the feature representation of the vectors using a lightweight CNN, and finally uses nonmaximal suppression to obtain the aircraft's bbox. Compared to the traditional CFF-based methods, CGC-NET significantly reduces the computational complexity at the model level by introducing a lightweight network structure. However, using a sequence of circumferential pixel gray values as input limits the model's ability to extract other texture features or deep semantic features of the object. CGC-NET still faces the challenge of computational cost when extracting sequences of circular pixels with different radii. As the radius increases, the number of pixels to be processed grows in square steps, which may lead to a sharp increase in the demand for computational resources.

In conclusion, despite significant advancements in aircraft detection technology over the past few decades, enhancing networks' ability to comprehend and leverage physical knowledge features remains a critical direction for future research. The method proposed in this article not only retains the powerful feature extraction capabilities of end-to-end networks but also learns the physical knowledge features of aircraft. It provides a new avenue for enhancing aircraft detection in RS images and opens up new possibilities for integrating physical knowledge features into deep learning models.

III. Method

In this section, we first introduce the overall structure of the proposed PKG-Net and describe the computation of the CG feature representation. Subsequently, we introduce the three key modules of PKG-Net: MS-CFFM, ACSM, and HAFFM.

A. Network Structure

The overall architecture of our PKG-Net is shown in Fig. 4, where the input image is processed by two distinct branches. The first branch utilizes the Swin Transformer to extract features. The second branch is dedicated to extracting the multiscale CG feature of the image through MS-CFFM, which

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then supervises the output of stage 3 from the first branch using ACSM. Subsequently, the features extracted by the backbone network are fused through HAFFM. Finally, the network performs aircraft classification and localization through its head. In Fig. 4, the paths in the gray dashed box represent are only involved in training and not in inferencing.

1) Feature Extraction Path: For a given RS image $X \in \mathbb{R}^{H\times W\times 3}$, the model utilizes the Swin Transformer as the backbone network. Initially, the data is divided into multiple small patches, each measuring 8×8 pixels. Subsequently, a linear embedding layer projects each patch into a C-dimensional space, forming a 3-D feature map. Finally, multiple hierarchical Transformer layers sequentially process the feature map, and within each Transformer layer, a moving window mechanism is applied, meaning the network computes self-attention within small local windows only. The network processes the feature map at different stages with varying resolutions to capture multiscale features. Specifically, the network has four feature extraction stages, with the output of each stage defined as S_n, where $n = 1, 2, 3, 4$. The resolution of the output feature map at stage n is $H/(2^{n+1}) \times W/(2^{n+1})$, and the dimension is $2^{n+1}C$.

2) Knowledge-Guided Path: For a given RS image $X \in \mathbb{R}^{H\times W\times 3}$, we first utilize MS-CFFM to manually generate CG feature maps $X_S \in \mathbb{R}^{H_1\times W_1\times 1}$ from the data. In addition, the feature map S_3 is input into ACSM, where channels with significant contributions are selected for fusion, resulting in a feature map $X_D \in \mathbb{R}^{H_1\times W_1\times 1}$. In the final stage, we design a specific contrastive loss function to regulate X_S and X_D, ensuring the network effectively learns the CG feature at stage 3.

3) Feature Fusion Path: After completing the four encoding stages of the backbone, we obtain the feature maps S_n. In our experiments, the feature maps (S_1, S_2, S_3) from the last three stages are fed into the feature fusion pathway. We use the widely recognized YOLOXPAFPN [32] as the foundational structure for feature fusion and apply custom processing to the feature maps from different stages. Specifically, we employ a coordinate attention module (CAM) [33] to improve the feature map output from stage 3, enhancing the neural network's cognition of the aircraft's spatial position by encoding location information. Channel self-correlation matrix (CSCM) is used to process the feature maps output from other stages, amplifying critical features by adjusting the channel features' weights. Furthermore, we innovatively replace the bottom-up cross-stage partial (CSP) layers [32] in the feature pyramid with CSP_dilated convolution (DC) layers. Following this comprehensive set of modifications and series of processes, the enhanced multiscale feature maps are directed to the decoder, which then generates the final detection predictions.

Three paths are used simultaneously for training and inference. The knowledge-guided paths are not involved in the computation when inferencing.

B. Multiscale Circular Frequency Filter Module

To reduce the time cost of extracting CG features pixel by pixel and to overcome the limitation that a single circular radius can only capture CG features at a single scale, we design multiscale convolutional kernels to generate feature maps at four scales. These are then linearly fused to produce the final CG feature map. Below, we will introduce the design principles of the convolutional kernels and the structure of MS-CFFM.

1) Fast Convolutional Kernels Based on CG Feature: Assuming the pixel point (i, j) as the center, we uniformly extract a 1-D array $p^n_{i,j,r}$ $(n \in [0, N-1])$ in the clockwise direction along the circumference of radius r. Perform a 1-D DFT on this array in accordance with (1) to obtain the amplitude response $|X[k]|^2$

$$X[k] = \sum_{n=0}^{N-1} p^n_{i,j,r} \cdot e^{-\frac{i2\pi}{N}kn} \tag{1}$$

$$|X[k]|^2 = \left(\sum_{k=0}^{N-1} p^k_{i,j,r} \cos\frac{2\pi}{N}kn\right)^2 + \left(\sum_{k=0}^{N-1} p^k_{i,j,r} \sin\frac{2\pi}{N}kn\right)^2. \tag{2}$$

Here, $X[k]$ denotes the frequency-domain signal, with k being the frequency index ranging from 0 to $N-1$. The term $e^{-(i2\pi/N)kn}$ is the rotation factor used to transform the time-domain signal into the frequency domain.

DFT actually measures the cumulative similarity between the time-domain signal and a rotation factor of a specific frequency across the entire time series. If the time-domain signal contains a significant component at a specific frequency, then the corresponding DFT output $X[k]$ will have a larger value.

For instance, considering $k = 1$, the calculation formula for $X[k]$ can be represented as (3). Following Euler's formula, the rotation factor can be articulated as a combination of sine and cosine functions, as shown below:

$$X[1] = \sum_{n=0}^{N-1} p^n_{i,j,r} \cdot e^{-\frac{i2\pi}{N}n} \tag{3}$$

$$e^{-\frac{i2\pi}{N}n} = \cos\left(-\frac{2\pi}{N}n\right) + i\sin\left(-\frac{2\pi}{N}n\right). \tag{4}$$

Multiplying the time-domain signal by the rotation factor and summing the results actually computes the similarity of the time-domain signal to two sine waves with a period of N, specifically $\cos((2\pi n/N))$ and $\sin((2\pi n/N))$.

The grayscale value transformation of the pixel point sequence sampled around the circumference with the airplane's center as the center exhibits a pattern of four peaks and four valleys. This implies the presence of a frequency component with a period of $N/4$ in the sequence. Consequently, at the frequency index $k = 4$, the amplitude response of the DFT is maximal, indicating that this is the most significant frequency component within the sequence. Therefore, $k = 4$ is chosen in (1) and (2).

Due to the properties of the Fourier transform, changes in the starting position of the circular pixel point sequence $p^n_{i,j,r}$ equate to a phase shift in the sampling sequence, meaning the amplitude response of its Fourier transform remains unaffected. This implies that for aircraft in different orientations, a large amplitude response can still be achieved,

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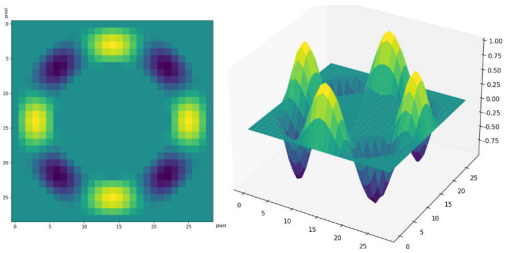


Fig. 5. Planar schematic and 3-D schematic of the convolution verification section.

demonstrating the rotational invariance of the CG feature. Direct circular sampling for every pixel in the image would make the computation exceedingly complex. To simplify this process, we constructed an image convolution template K, intended for fast computation of (2)

$$K(r,\varphi) = \sin\left(\frac{\pi}{R - r_f}\left(r - r_f\right)\right)e^{ik\varphi}$$
$$\text{where } k = 4,\ R = \frac{\text{conv_size} - 1}{2}. \tag{5}$$

Here, r_f represents the flat radius, conv_size is the size of the convolution kernel, and φ is an angle value within the range from 0 to 2π. Equation (5) is composed of a sine function and the Fourier basis function $e^{ik\varphi}$ (with $k = 4$), facilitating the convolution over the image. The planar and 3-D schematics of the real part of the convolution template are shown in Fig. 5, respectively, where conv_size is chosen to be 29, and r_f is selected as 6.

2) Detailed Process for MS-CFFM: Due to the uncertainty in the scale of aircraft in RS images, convolution kernels with a single sampling radius only respond significantly to aircraft within a specific scale range, while showing poor responses to aircraft of other scales. This leads to insufficient extraction of aircraft features. To overcome this issue, this article proposes the MS-CFFM, aimed at effectively extracting features of aircraft at different scales. By using circular frequency convolution kernels with different sampling radii on the same RS image, multiple feature maps are generated. These feature maps are linearly fused in a predetermined ratio to form a comprehensive final feature representation that reflects features of different scales.

In the experiments, we first conducted a detailed statistical analysis of the aircraft's pixel area. Based on the obtained data, four different sizes of convolution kernels were designed (3×3, 10×10, 29×29, and 78×78, respectively). Considering that smaller kernels are capable of capturing more detailed feature information, we adopted a weight ratio of 2:1:1:1 for the linear fusion of these four feature maps, to extract the final comprehensive feature map.

C. Adaptive Channel Selection Module

In previous supervised learning models, the loss is usually computed in all feature channels. If the model is too focused on using all channels to learn CG features, the model's ability to extract other texture features and semantic features of the image will be limited. Therefore, we need to limit the number of channels for learning CG features to find a perfect balance between the model learning CG features and autonomously extracting other features. To address this issue while preserving the model's original feature extraction capabilities, we propose the ACSM. ACSM is aimed at enabling the network to autonomously decide which channels to activate for learning CG feature. This approach effectively learns CG features while minimally impacting the model's original feature extraction capabilities.

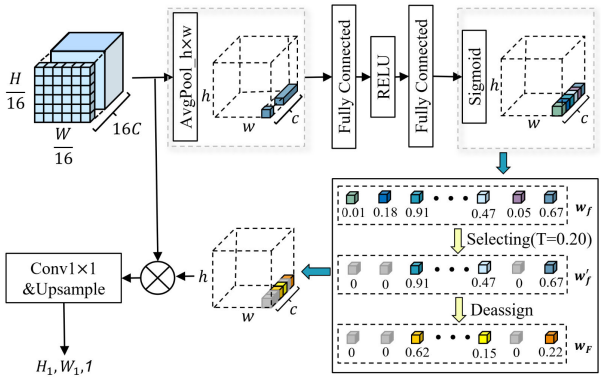


Fig. 6. Structure of ACSM. First, the input feature map is pooled globally, and then the activation values for each channel are generated. Based on the threshold T, channels with activation values greater than T (colored cubes) are selected and retained, while channels with activation values less than T (gray cubes) are not activated and ignored. Eventually, the retained channels are weighted, convoluted, and upsampled to generate a weighted feature map for subsequent stages.

The structure of ACSM is shown in Fig. 6. It adaptively generates activation values for each channel based on the input feature map. These activation values are used to rank the channels by importance. The channels are then pruned according to a predetermined threshold, discarding channels with activation values below the threshold while retaining those with activation values above or equal to the threshold. For these retained channels, the system calculates activation weights and adjusts their sum to 1. Finally, the input feature map is reweighted according to the activation weights to form a new feature map. This feature map is processed through a convolutional layer and upsampled to reduce its channel dimension to 1 while enhancing its resolution.

To adaptively learn the weight of each channel, we draw inspiration from the SE module to model the relationships among channels. First, the feature map $S_3 \in \mathbb{R}^{(H/16)\times(W/16)\times 16C}$ is compressed into a feature vector through global average pooling. Then, by employing fully connected layers and nonlinear activation functions, we learn to generate an activation value vector for the channels

$$w_f = \sigma\left[W_2\delta\left(W_1 X_{\text{pool}}\right)\right] \tag{6}$$

$$X_{\text{pool}} = \frac{1}{H \times W}\sum_{i=1}^{H}\sum_{j=1}^{W} S_3(i,j). \tag{7}$$

Here, σ and δ represent the activation functions ReLU and Sigmoid, respectively. W_1 and W_2 represent the weights of the first and second fully connected layers, respectively. w_f is the activation value vector for the channels; $X_{\text{pool}} \in \mathbb{R}^C$ represents

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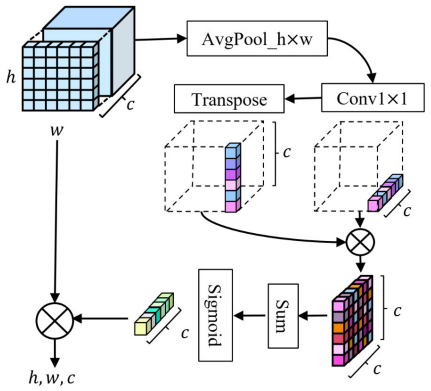


Fig. 7. Structure of CSCM.

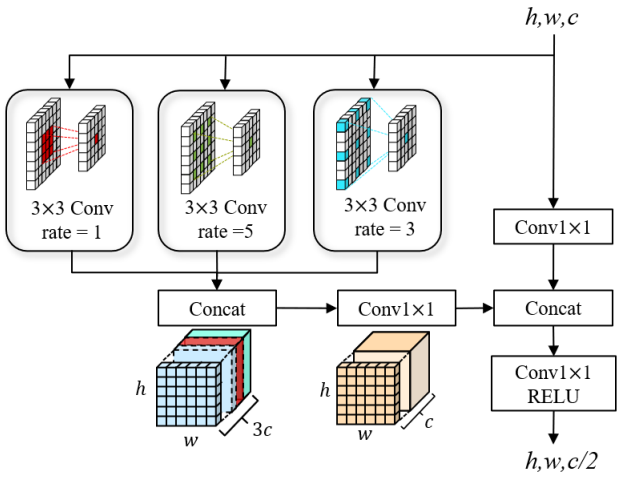


Fig. 8. Structure of CSP_DC.

the result compressed in its spatial dimensions $H \times W$ through the global average pooling operation.

Let $w_f = [p_1, p_2, \ldots, p_C]$ represent the activation value vector for the channels. According to the threshold T (which is set to 0.20 in the experiment), decide whether to set p_i to 0 ($i \in [1, C]$), resulting in a new activation value vector $w'_f = [p'_1, p'_2, \ldots, p'_C]$. Then, according to (9), normalize the sum of all elements in the activation value vector to 1, obtaining the activation weights $w_F = [P_1, P_2, \ldots, P_C]$

$$p'_i = \begin{cases} 0, & \text{if } p_i \le T \\ p_i, & \text{otherwise} \end{cases} \tag{8}$$

$$P_i = \frac{e^{p'_i}}{\sum_{i=1}^{C} e^{p'_i}}. \tag{9}$$

Finally, the weighted feature X_D^w is generated by multiplying w_F and S_3. This weighted feature is then processed through a convolutional layer to adjust its channel number to 1, and upsampled to match the resolution of X_S

$$X_D^w = w_F S_3 \tag{10}$$

$$X_D = \text{UpSample}\big(\text{conv}\big(X_D^w\big)\big). \tag{11}$$

The weighted feature X_D is supervised to fit the CG feature X_S. Following this, the network autonomously adjusts the activation weights of the channels within the feature map in response to feedback from the loss function, thereby augmenting or diminishing their influence. ACSM focuses the network more on the channels that are essential for learning CG features, thereby significantly optimizing the efficiency of the learning process.

D. Hybrid Attention Feature Fusion Module

To enhance the model's focus on aircraft location information, emphasize the intrinsic connections and distinctions between different channels of the feature map, and capture more comprehensive contextual information, we make customized improvements to PAFPN by introducing CAM, CSCM, and CSP_DC. Specifically, we incorporate the CAM module to enhance the expressiveness of the feature maps output by stage 3 and design the CSCM algorithm for feature maps of stage 2 and stage 4. In addition, we innovatively replaced the top-down CSP layers in the feature pyramid with CSP_DC layers. Fig. 4 shows the structure of the HAFFM.

1) Channel Self-Correlation Matrix: The Swin Transformer mainly utilizes a single-channel-based self-attention mechanism during the feature extraction process, which shows certain limitations in terms of inter-channel interaction. To address this issue, we propose the CSCM. This algorithm, by learning the self-correlation weights between channels, can redistribute the focus on different channels, enabling the model to selectively concentrate on specific channels and effectively suppress irrelevant or redundant information. The exact flow of the algorithm is shown in Fig. 7. Initially, a global pooling operation is performed on the input feature map to extract the average information of each channel feature map. Then, channel weights are encoded through a 1×1 convolution layer. Furthermore, a self-correlation matrix is computed along the channel dimension, and the sum of the elements in each column is calculated. Then, weights are recalculated through a sigmoid layer. Finally, these weights are multiplied by the input feature map to achieve adaptive weighting of different channels, thereby optimizing the model's perception of channel interconnectivity.

2) Cross-Stage Partial_Dilated Convolution: To enhance the model's ability to capture image details and the global perception of features, we design the CSP_DC layer. As shown in Fig. 8, the input feature map is first processed in parallel by three 3×3 convolution layers with different dilation rates to synchronously capture multiscale contextual information. Then, these feature maps processed by DCs are concatenated together, followed by dimension reduction through a 1×1 convolution layer. Finally, the module utilizes a residual connection to combine the original input feature with the compressed feature and introduces nonlinearity through the ReLU activation function, outputting the final feature map.

E. Loss Functions

The overall loss used to optimize our network is provided by (12), which includes the loss of circular frequency features and the loss of detection

$$L_{\text{total}} = L_{\text{det}} + L_{\text{CFF}}. \tag{12}$$

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The loss function of detection is based on a combination of the classification loss and the regression loss. It can be expressed as follows:

$$L_{\text{det}} = L_{\text{cls}} + \lambda L_{\text{reg}}. \tag{13}$$

Here, L_{cls} represents the classification loss, L_{reg} represents the regression loss, and λ is a hyperparameter representing the balancing coefficient for the regression loss, generally set to 5.0. L_{cls} utilizes the binary cross-entropy loss [34], while L_{reg} utilizes the IoU loss function [35]

$$L_{\text{cls}} = -\sum_{i=1}^{n} (t_i \log(p_i) + (1 - t_i) \log(1 - p_i)) \tag{14}$$

$$L_{\text{reg}} = -\log\left(\text{IoU}\left(B_{gt}, B_{\text{pred}}\right)\right). \tag{15}$$

Here, t_i represents the one-hot vector of the true box category, p_i represents the one-hot vector of the predicted box category, B_{gt} represents the coordinates of the true box, and B_{pred} represents the coordinates of the predicted box.

L_{CFF} combines two mean squared error (MSE) components: the first component focuses on the circular frequency features within the object box, and the second component is dedicated to the background areas

$$L_{\text{CFF}} = \alpha \text{MSE}\left(X_D^{\text{bbox}}, X_S^{\text{bbox}}\right) + (1 - \alpha)\text{MSE}\left(X_D^{\text{background}}, X_S^{\text{background}}\right). \tag{16}$$

Here, X_D^{bbox} and X_S^{bbox} represent the bounding box areas extracted from the feature maps X_D and X_S, respectively, $X_D^{\text{background}}$ and $X_S^{\text{background}}$ represent the background areas extracted from the feature maps X_D and X_S, respectively. α is a weight parameter, varying from 0 to 1, that is utilized to adjust the contribution proportion of the loss within the bounding box areas.

IV. Experiments

In this section, we introduce the datasets utilized in the experiments, followed by detailed descriptions of the experimental setup. Next, we conduct ablation experiments to analyze various modules of PKG-Net both qualitatively and quantitatively. Finally, we present the results of different methods on each dataset.

A. Datasets

1) RSOD Dataset: The RSOD Dataset is a public RS image dataset for object detection. It includes 446 images with 4993 airplanes, 189 images with 191 playgrounds, 176 images with 180 overpasses, and 165 images with 1586 oil barrels. In our study, we only utilized images containing airplanes. For the experiment, 20% of the images were used as a validation set, 10% as a test set, and the remaining 70% for training. The training set was constructed by randomly sampling from these 70% of images, ensuring a varied representation of object coverage. This includes images with all objects, as well as those containing one-seventh, three-sevenths, and five-sevenths of the total number of objects.

2) UCAS- AOD Dataset: The UCAS-AOD dataset comprises 1510 aerial images from Google Earth, covering two categories: 510 images of cars and 1000 images of airplanes. In our experiments, we adopt the same division strategy as used with the RSOD dataset, splitting the dataset into training, validation, and test sets.

B. Implementation Details

1) Training Settings: Our network was built under the MMDetection framework. We employed the AdamW optimizer with a weight decay of $5e^{-4}$ for training the model. The learning rate was set at $4e^{-4}$, using a Warmup strategy to gradually increase the rate at the beginning of training, followed by a decrease through the CosineAnnealingLR strategy, and finally maintaining a fixed rate in the final stages of training. All experiments were conducted on an NVIDIA Geforce RTX 3090 24-GB GPU. The batch size was 8, and the maximum training epoch was 100. For all input images, their sizes were adjusted to 800×800.

2) Evaluation Index: We have adopted the COCO [36] evaluation metrics to assess the performance of our model. We report the average precision (AP) across all Intersection over Union (IoU) thresholds, AP_{50} at IoU 0.5, and AP_{75} at IoU 0.75. We also measure AP for different object sizes, including AP for small objects (pixel area $<32^2$) as AP_S, AP for medium objects ($32^2 \leq$ pixel area $\leq 96^2$) as AP_M, and AP for large objects (pixel area $>96^2$) as AP_L. Considering the nature of small objects, a minor pixel deviation can lead to significant fluctuation in IoU. An IoU of 50% between the predicted bounding boxes and the ground-truth labels is considered a reasonable prediction, thus AP_{50} may serve as a more effective and fair metric.

C. Ablation Study

To assess the efficacy of the proposed network architecture and three key modules, we utilize the Swin Transformer as the baseline network for conducting ablation experiments and module replacement trials on the RSOD dataset. We meticulously explore various hyperparameters to identify the optimal configuration that maximizes performance. Fig. 9 shows the visualization of CG feature learned at different stages. Fig. 10 shows the results of module replacement. Fig. 11 presents hyperparameter experiment results, and Fig. 12 displays module ablation results. In our PKG-Net, Swin is used in the backbone network with "Tiny" configuration (i.e., the hidden size $C_1 = 96$, the window size is 8, the number of layers corresponding to each stage is {2, 2, 6, 2}, and the number of heads corresponding to each layer is {3, 6, 12, 24}). Moreover, we initialize the backbone with ImageNet pretrained weights and employ yoloxhead for the detection head.

1) Effect of Learning CG Feature at Different Stages: We employ various stages of Swin to learn CG features and conduct a quantitative evaluation of the effectiveness of different stages in enhancing detection accuracy. The results are shown in Table I. Utilizing partial channels of stage 3 to learn CG features significantly enhances the detection performance

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TABLE I
ABLATION EXPERIMENTS ON LEARNING CG FEATURE AT DIFFERENT SWIN STAGES ON THE RSOD DATASET

Training Set Ratio	Stage			Evaluation Index					
	stage2	stage3	stage4	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l
1/7	✓	-	-	53.6	93.0	57.4	37.2	57.5	65.1
	-	✓	-	**55.5**	**93.4**	**61.4**	**38.1**	**59.8**	66.3
	-	-	✓	53.3	92.9	54.7	34.4	56.3	**68.4**
3/7	✓	-	-	61.1	95.3	73.3	44.2	64.4	75.5
	-	✓	-	**62.6**	**96.1**	**73.7**	**45.5**	**64.8**	**77.7**
	-	-	✓	58.1	95.0	69.5	40.6	61.1	76.9
5/7	✓	-	-	63.2	96.6	77.8	47.3	67	77.8
	-	✓	-	**65.1**	**97.2**	**79.9**	**48.1**	**67.9**	**79.6**
	-	-	✓	62	96.3	74.3	44.3	65	78.9
1	✓	-	-	67.0	97.0	83.4	49.7	69.9	79.7
	-	✓	-	**67.1**	**97.3**	**84.4**	**50.9**	**70.1**	**81.3**
	-	-	✓	66.5	96.8	80.9	48.8	69.2	80.7

TABLE II
ABLATION EXPERIMENT OF THE PROPOSED MODULES ON THE RSOD DATASET

Training Set Ratio	Moudules			Evaluation Index					
	MS-CFFM	ACSM	HAFFM	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l
1/7	-	-	-	48.3	89.1	46.3	34.1	52.6	47.3
	✓	-	-	53.1	92.1	58.9	37.3	56.7	57.7
	✓	✓	-	53.7	92.9	60.5	37.8	58.6	62.7
	✓	✓	✓	**55.5**	**93.4**	**61.4**	**38.1**	**59.8**	**66.3**
3/7	-	-	-	55.3	94.4	60.6	41.5	57.9	62
	✓	-	-	61.6	95.1	71.7	43.6	63.6	74.8
	✓	✓	-	61.8	95.9	72.6	44.8	64.2	75.9
	✓	✓	✓	**62.6**	**96.1**	**73.7**	**45.5**	**64.8**	**77.7**
5/7	-	-	-	60.5	95.9	71.3	43.8	63.6	66.5
	✓	-	-	64.2	96.5	77	44.9	67.1	75.5
	✓	✓	-	64.8	96.9	78.1	46.7	67.3	77.5
	✓	✓	✓	**65.1**	**97.2**	**79.9**	**48.1**	**67.9**	**79.6**
1	-	-	-	62.4	96.7	75.1	46.2	65.6	67
	✓	-	-	65.4	96.9	83.4	49.8	69.7	78.6
	✓	✓	-	66.4	97.1	83.7	50.6	69.9	80.3
	✓	✓	✓	**67.1**	**97.3**	**84.4**	**50.9**	**70.1**	**81.3**

of this method. Especially under the condition where the sample ratio is only one-seventh (the third, fourth, and fifth rows of Table I), compared to stage 2 and stage 4, stage 3 increases the AP by 1.9% and 2.2%, respectively. The feature maps of stage 3 provide the best balance between details and semantic information, which is the advantage of learning CG features. The focus on aircraft localization not only enriches low-level and mid-level feature information but also enhances the model's understanding of the context between the aircraft and the background. More intuitively, Fig. 9 presents the heat maps of the features outputted in stage 2 after learning at three distinct stages. Specifically, stage 2 (the second row of Fig. 9) reveals relatively dispersed activation points, indicating that the network captures a rough estimate of the aircraft's center at an early stage, but has not entirely filtered out the surrounding background noise. Stage 4 (the fourth row of Fig. 9) displays lower feature activation values, which could be attributed to the model capturing more contextual information at deeper network stages, rather than focusing solely on central features. Stage 3 (the third row of Fig. 9) shows more concentrated and pronounced feature activation around the aircraft's center, which might be because stage 3 provides higher-level abstract features that may include spatial relationships and finer object details, contributing to a more accurate capture of the aircraft's central position.

2) Effect of MS-CFFM: MS-CFFM extracts and fuses multiscale CG features to guide the model learning, which greatly improves the network's ability to locate aircraft at different scales. This module extracts a stable feature of aircraft that aligns with human cognition and is adaptable to various environmental changes, enabling the model to precisely identify aircraft positions under diverse angles and backgrounds. In Table II, compared with the baseline, AP_{50} is improved by 3%, 0.7%, 0.6%, and 0.2% for the training set ratios of 1/7, 3/7, 5/7, and 1, respectively. This trend may be due to the strong feature extraction ability of the Transformer when the sample size is sufficient, which limits the upper limit of the CG feature in improving the performance. The visualization heatmap [Fig. 12(d)] shows that the activation areas near small- and medium-sized aircraft are more concentrated in the center of the aircraft than those extracted by the baseline. In addition, the activation areas of

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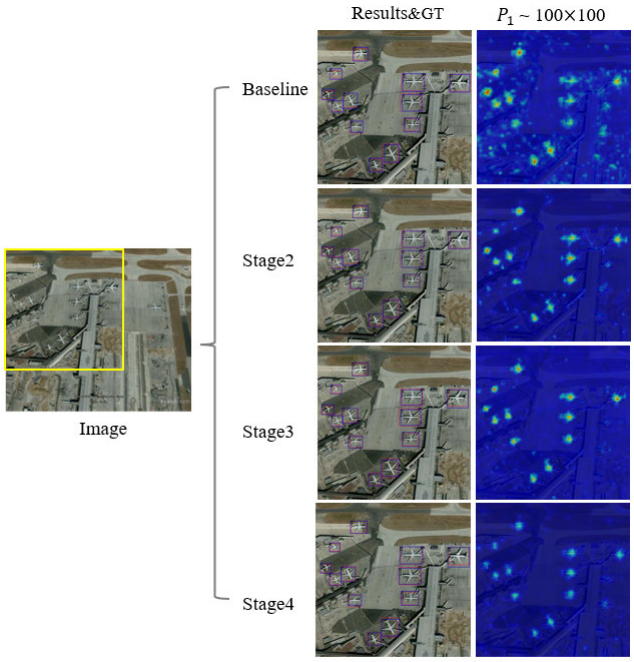


Fig. 9. Detection results and activation maps of learning CG feature at different stages.

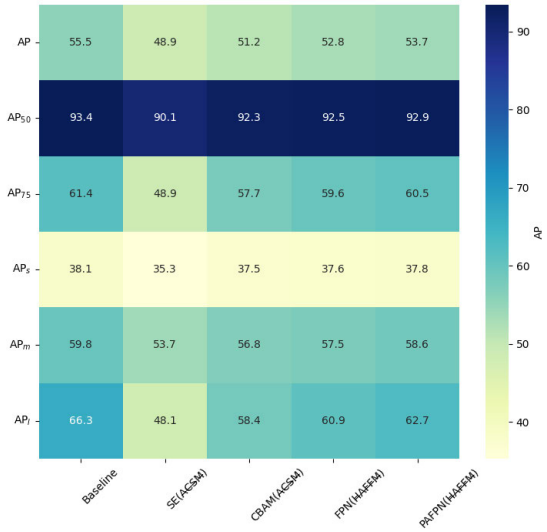


Fig. 10. Module replacement experiments with PKG-Net components on the RSOD dataset at a 1/7 training sample ratio. We replaced ACSM using SEM and CBAM, and HAFFM using FPN and PAFPN, respectively. The darker the color, the higher the accuracy.

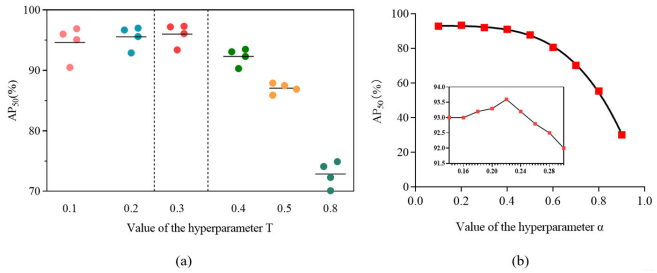


Fig. 11. Effect of different values of hyperparameters on model performance. (a) Hyperparameter T in the ACSM increases, the model accuracy first increases and then decreases, with an extreme value at T of 0.3. (b) Effect of the hyperparameter α in the loss function on the model accuracy. The detection accuracy is highest when α takes the value of 0.22.

large-scale aircraft are slightly dimmed. It may be attributed to the fact that MS-CFFM uses multiscale convolution kernels to extract feature maps and generates a comprehensive feature representation through linear fusion that assigns greater weight to smaller scales. This module weakens the network's attention to large-scale aircraft and balances the model's allocation of attention to aircraft of various scales.

3) Effect of ACSM: ACSM selectively activates channels to learn CG features, helping the model balance feature learning. In Table II, compared with detection results of the baseline (+MS-CFFM) (the fourth row of Table II), remarkable improvements on all indicators are acquired by the baseline (+MS-CFFM + ACSM) (the fifth, ninth, thirteenth, and seventeenth rows of Table II). Specifically, scores of AP, AP_{50}, AP_{75}, AP_s, AP_m, and AP_l are 0.6%, 0.8%, 1.6%, 0.5%, 1.9%, and 5.0% superior to those of the baseline (+MS-CFFM) for the training set ratios of 1/7, respectively. A more intuitive visualization of the feature heatmap in Fig. 12(f), where one can observe a clearer separation between densely parked small aircraft than in the baseline (+MS-CFFM). In addition, the uniformity of the background is significantly improved. This may be due to the pruning and activation weight adjustment of ACSM, which reduces the influence of noise channels and improves the response of channels important to aircraft characteristics. To further exemplify the effectiveness of ACSM, we experiment with replacing ACSM with a squeeze-and-excitation (SE) module and a convolutional block attention module (CBAM). As shown in Fig. 10, ACSM outperforms AP values across different thresholds and sizes. This might be because SE and CBAM learn CG features across all channels, which leads to overfitting CG features and underfitting other features.

4) Effect of HAFFM: HAFFM focuses on the connection between channels and the perception of global features, which enhances the expressive ability of feature maps. In comparison to the detection results of the baseline (+MS-CFFM + ACSM) and the baseline (+MS-CFFM + ACSM + HAFPN) (the sixth, tenth, fourteenth, and eighteenth rows of Table II), scores of AP, AP_{50}, AP_{75}, AP_s, AP_m, and AP_l achieved by the baseline (+MS-CFFM + ACSM + HAFPN) are 1.8%, 0.5%, 0.9%, 0.3%, 1.2%, and 3.6% higher than those of the baseline (+MS-CFFM + ACSM) for the training set ratios of 1/7. It fully verifies the effectiveness of HAFFM in refining important aircraft features and understanding global context information. The visualization result [Fig. 12(h)] shows that the activation areas for aircraft of all sizes become brighter. Especially in scenarios with tightly arranged aircraft [the third row in Fig. 12(g)], the model offers clearer segmentation and more precise positioning. Through employing refined feature extraction, selection, and fusion strategies, HAFFM emphasizes the intrinsic connections between different channels and coordinates, provides richer contextual information, and improves the hierarchy and consistency of aircraft features. Similarly, we use feature pyramid network (FPN) and PAFPN instead of HAFFM, and the results are shown in Fig. 10. Unlike conventional pyramid structures that focus on

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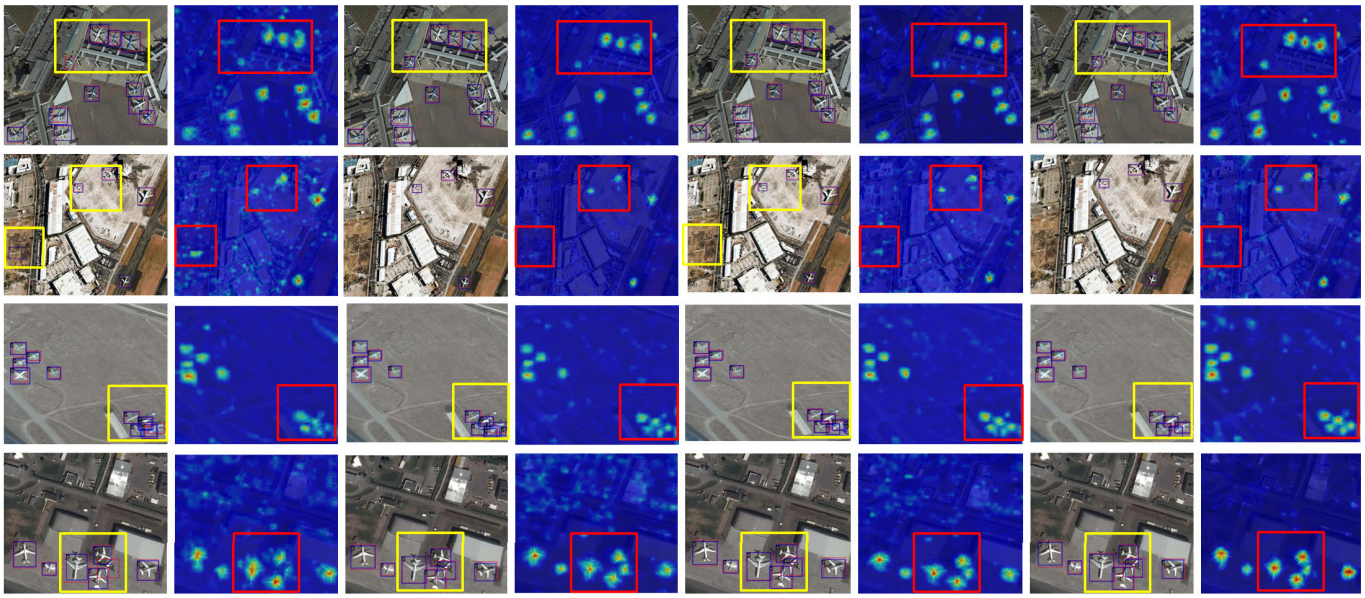


Fig. 12. Detection results and activation maps produced by our method equipped with the proposed submodules progressively. (a) and (b) Baseline. (c) and (d) Baseline (+MS-CFFM). (e) and (f) Baseline (+MS-CFFM+ACSM). (g) and (h) PKG-Net (Ours).

connections between levels, HAFFM enhances interactions in both spatial and channel dimensions, improving the precision of small and medium targets by 0.5% and 2.3% compared to FPN, respectively.

5) Hyperparameter Analysis: To evaluate the impact of hyperparameters on model performance, we quantitatively analyze the specific impact of the choice of the threshold T in the ACSM and the α parameter in the loss function on the model accuracy.

As shown in Fig. 11(a), the hyperparameter T set to 0.2 is an equilibrium point at which the model may be neither too weak nor overemphasized on the CG feature, thus avoiding under- or overfitting. Fig. 11(b) shows the detection accuracy of hyperparameter α when the experiment is within the range of 0.1–0.9, with intervals of 0.1. A finer-grained division is then conducted, with values ranging from 0.14 to 0.3, in increments of 0.22. The aircraft that is the object of detection tends to be small relative to the entire image, while the background contains most of the pixels. α set to 0.22 provides a delicate balance that ensures that the model learns features adequately on both the aircraft and the background.

D. Comparison With Other Methods

We compare our method with several prevalent object detection methods to underscore its competitive edge, including RetinaNet, Faster R-CNN, Cascade R-CNN, and DINO, each harnessing the ResNet-101 as their backbone. Alongside these, Yolox and Yolo-ms are considered, followed by DDQ DETR and Co-Deformable-DETR, RT-DETR, which are both built on the Swin Transformer architecture. To complete our evaluation lineup, we introduced RetinaNet and Mask R-CNN with Swinv2-T as the backbone network. In our comparison experiments, all the methods employing ResNet-101 as the backbone network used weights obtained from pretraining on the ImageNet 1K dataset. Comparatively, the Swin-T and Swinv2-T-based methods used weights pretrained on the larger ImageNet 22K dataset.

1) Results on the RSOD Dataset: Tables III and IV present the numerical results for each detection method. The results indicate that our proposed PKG-Net surpasses other methods in both AP and AP_{50} across various training set ratios. Faster R-CNN employs a region proposal network and a two-stage detection mechanism that effectively utilizes features from larger regions. Cascade R-CNN adopts a multitier cascading strategy, where each tier refines the output of the previous one. This strategy is particularly effective for detecting large-size objects in complex scenes, as it progressively eliminates biases during detection. The data show that Faster R-CNN outperforms PKG-Net by 0.7% in AP_l at a 1/7 training set ratio. Cascade R-CNN also outperforms PKG-Net by 0.7% on AP_l at a 3/7 training set ratio. This performance gap may arise from our MS-CFFM design's fusion strategy, which favors smaller-sized objects and may not adequately consider the features of larger objects. When the training sample ratio is at 5/7 or a full sample, our PKG-Net outstrips all other comparative approaches in overall performance. At each training set ratio, compared to Yolox, our PKG-Net achieves an approximate increase of 16.3%, 4.2%, 1.3%, and 2.4% in AP, respectively. Compared to the advanced RT-DETR, our PKG-Net achieves an approximate increase of 6.7%, 1.8%, 0.7%, and 0.5% in AP_{50}, respectively.

Fig. 13 presents the prediction results of different methods in aircraft detection tasks. In the first scenario, RetinaNet, both Mask R-CNN and YOLOX, encounter bounding box overlap issues when dealing with densely arranged small aircraft. Although other methods successfully identified most objects, their bounding boxes were not as tightly fit as those

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TABLE III
DETECTION RESULTS OF DIFFERENT METHODS ON THE RSOD DATASET WITH TRAINING SET AT RATIOS OF 1/7 AND 3/7

Training Set Ratio		1/7						3/7					
Method	Backbone	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l
RetinaNet [37]	ResNet-101 [38]	27.9	55.7	25.3	10.7	31.4	47.6	46.7	81.8	49.2	24.4	50.5	67.8
Faster R-CNN [14]		45.8	85.1	45.1	24.8	49.3	**67.0**	55.5	91.8	62.7	27.8	62.2	70.6
Cascade R-CNN [39]		45.4	85.5	45.1	22.3	49.4	64.6	56.8	90.7	67.0	36.6	60.4	**78.4**
DINO [40]		47.5	85.2	40.5	25.1	52.4	59.6	58.6	93.1	69.1	35.7	60.5	74.8
Yolox [32]	CSPDarkNet	39.2	78.8	34.4	15.7	44.2	47.2	58.4	93.5	69.6	39.6	61.7	69.7
Yolo-ms [41]		45.2	82.1	38.1	17.8	47.3	50.9	59.4	93.8	70.1	42.7	61.9	71.6
DDQ DETR [42]	Swin-T [23]	47.1	84.7	48.0	26.5	51.8	60.0	57.7	93	67.7	33.1	61.8	73.6
Co-Defoemable-DETR [43]		30.7	54.3	30.9	19.2	33.2	45.1	49.0	78.3	56.3	35.5	51.4	68.7
RT-DETR [44]		48.1	86.7	42.1	26.4	52.1	61.8	60.1	94.3	70.9	41.5	62.3	75.7
RetinaNet	Swinv2-T [24]	32.9	73.3	21.1	9.2	36.8	50.0	46.8	86.6	45.4	24.6	51.5	63.7
Mask R-CNN [45]		46.6	87	44.6	27.6	51.5	54.5	56.5	90.7	66.2	37.9	59.7	71.4
PKG-Net	Swinv2-T	**55.5**	**93.4**	**61.4**	**38.1**	**59.8**	66.3	**62.6**	**96.1**	**73.7**	**45.5**	**64.8**	77.7

TABLE IV
DETECTION RESULTS OF DIFFERENT METHODS ON THE RSOD DATASET WITH TRAINING SET AT RATIOS OF 5/7 AND 1

Training Set Ratio		5/7						1					
Method	Backbone	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l
RetinaNet	ResNet-101	50.3	86.0	55.5	24.3	54.7	65.7	61.6	92.8	76.3	38.6	65.4	77.8
Faster R-CNN		56.4	93.9	62.3	30.8	62.8	75.7	61.8	94.4	73.1	36.6	67.4	77.2
Cascade R-CNN		59.2	94.2	66.9	37.9	62.4	77.8	63.5	95.4	77.3	43.5	66.0	79.6
DINO		62.1	95.7	78.9	45.7	65.1	77.9	63.8	96.7	77.8	43.8	68.9	80.1
Yolox	CSPDarkNet	63.8	95.0	78.5	44.8	67.0	76.8	64.7	95.8	79.6	46.3	67.6	77.8
Yolo-ms		64.5	95.9	79.0	46.6	67.7	78.7	65.9	96.2	80.8	47.9	68.1	79.2
DDQ DETR	Swin-T	61.0	95.5	72.2	44.9	64.1	79.5	62.3	96.5	75.4	40.2	66.0	83.0
Co-Defoemable-DETR		54.6	85.7	63.0	28.3	61.7	73.4	60.4	92.6	73.0	40.2	63.9	79.4
RT-DETR		63.8	96.5	79.1	46.2	66.7	78.7	66.8	96.8	81.1	48.2	68.9	80.3
RetinaNet	Swinv2-T	51.6	88.7	56.5	27.8	55.6	67.5	57.5	91.6	65.7	31.5	61.7	73.7
Mask R-CNN		55.7	93.3	61	37.1	58.8	72.3	62.9	95.6	75.6	43.0	65.9	77.2
PKG-Net	Swinv2-T	**65.1**	**97.2**	**79.9**	**48.1**	**67.9**	**79.6**	**67.1**	**97.3**	**84.4**	**50.9**	**70.1**	**81.3**

TABLE V
DETECTION RESULTS OF DIFFERENT METHODS ON THE UCAS-AOD DATASET WITH TRAINING SET AT RATIOS OF 3/7 AND 5/7

Training Set Ratio		1/7						3/7					
Method	Backbone	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l
RetinaNet	ResNet-101	35.9	67.7	36.6	0.9	35	57.8	48.5	82.5	5.5	1.1	49.3	68.0
Faster R-CNN		51.8	89.5	57.9	4.6	53.6	64.1	60.3	94.5	68.9	12.7	60.9	73.8
Cascade R-CNN		54.1	91.2	63.1	1.9	56.2	**72.8**	61.7	95.6	73.3	7.5	62.7	74.8
DINO		55.7	92.8	61.8	10.1	55.2	65.7	60.9	95.1	72.4	25.9	62.5	74.9
Yolox	CSPDarkNet	49.1	91.4	45.5	5.8	49.1	64.8	60.6	95.0	71.1	18.4	61.0	71.9
Yolo-ms		52.3	92.1	50.7	5.1	51.9	65.8	61.1	96.4	73.5	22.8	63.1	73.3
DDQ DETR	Swin-T	55.7	92.5	60.2	11.8	55.5	65.7	62.7	97.1	71.9	28.7	63.5	**76.8**
Co-Defoemable-DETR		37.4	61.3	40.7	10.1	38.2	43.9	56.3	85.4	67.7	24.7	57.5	57.0
RT-DETR		55.9	93.0	62.3	11.8	56.1	65.9	62.3	96.7	74.9	29.1	63.4	76.7
RetinaNet	Swinv2-T	46.1	86.6	43.6	4.5	46.9	54.4	56	92.5	62.9	9.5	56.7	64.8
Mask R-CNN		53.8	92.1	59.1	4.9	55.8	57.8	60.6	95.4	71.4	10.4	61.4	68.9
PKG-Net	Swinv2-T	**56.3**	**93.2**	**63.6**	**12.6**	**57.2**	67.1	**63.8**	**97.7**	**77.2**	**31.0**	**64.2**	75.2

produced by our method. In the second scenario, all methods except PKG-Net and DDQ-DETR exhibit some degree of object overlap or omission. In scenarios with densely arranged and complex backgrounds, the proposed PKG-Net method provides predictions that closely match the actual situation, demonstrating impressive performance.

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TABLE VI
DETECTION RESULTS OF DIFFERENT METHODS ON THE UCAS-AOD DATASET WITH TRAINING SET AT RATIOS OF 5/7 AND 1

Training Set Ratio		5/7						1					
Method	Backbone	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l
RetinaNet	ResNet-101	58.3	93.9	68.4	12.1	58.5	68.1	66.0	96.1	82.7	15.8	66.4	77.9
Faster R-CNN		62.5	95.6	76.6	14.4	63.1	73.6	68.2	97.6	84.8	23.9	68.4	78.9
Cascade R-CNN		65.3	95.9	80.3	16.8	65.5	76.9	68.8	97.8	86.0	35.8	68.5	78.5
DINO		65.1	96.7	80.6	32.5	65.1	76.4	68.1	97.9	86.3	45.9	67.4	77.2
Yolox	CSPDarkNet	65.2	97.2	79.9	29.0	65.4	75.2	67.3	98.2	86.4	29.4	67.5	75.1
Yolo-ms		65.5	97.5	80.8	33.9	65.8	76.1	67.5	98.2	86.6	34.5	67.6	76.8
DDQ DETR	Swin-T	65.6	98.3	80.9	34.5	64.2	77.4	67.8	98.4	85.0	44.4	68.0	79.1
Co-Defoemable-DETR		61.6	91.1	76.7	33.8	62.0	69.6	67.9	97.0	85.5	45.1	68.1	76.8
RT-DETR		65.7	97.8	81.7	35.1	64.9	77.2	68.9	98.2	87.0	46.1	68.3	78.7
RetinaNet	Swinv2-T	59.6	94.1	69.0	12.9	60.7	68.5	65.4	98.1	78.9	17.8	65.1	76.6
Mask R-CNN		64.7	95.4	78.6	18.1	65.4	75.5	68.5	97.7	84.3	24.8	68.6	78.5
PKG-Net	Swinv2-T	**65.8**	**98.5**	**82.9**	**35.6**	**66.0**	**77.5**	**69.6**	**98.6**	**87.1**	**46.4**	**68.7**	**80.1**

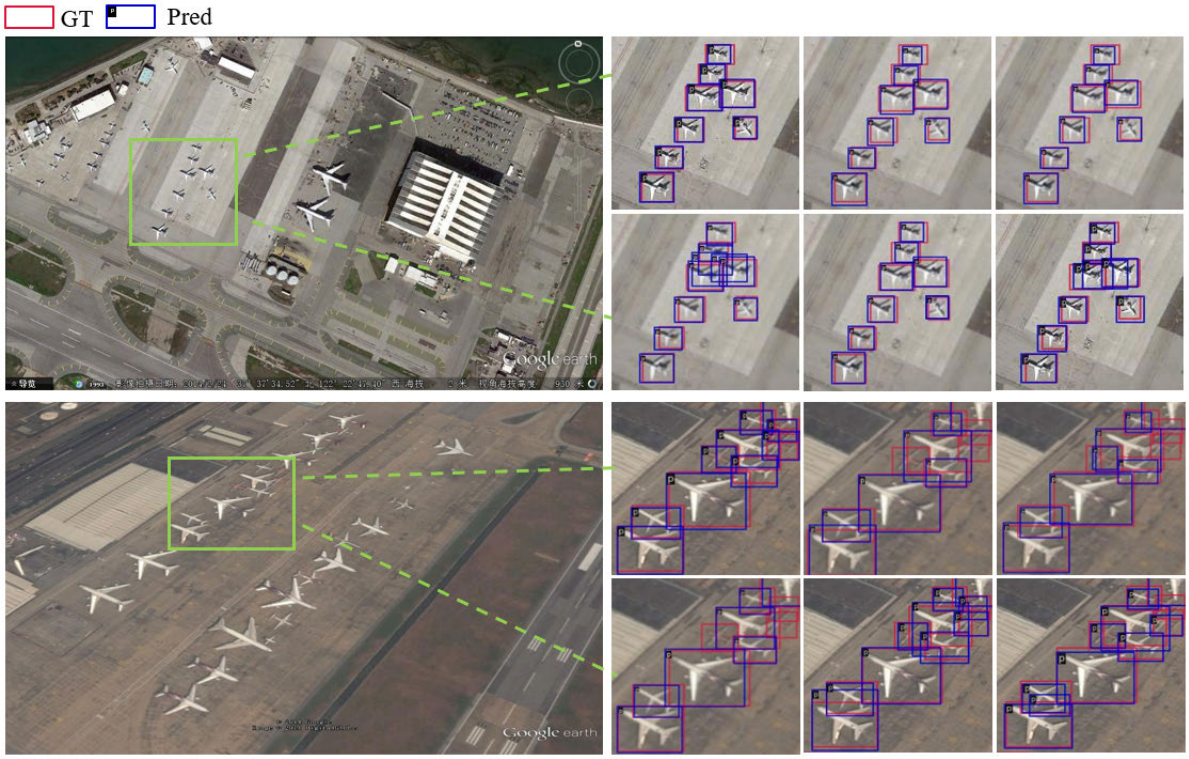


Fig. 13. Partially enlarged comparison of PKG-Net, Faster R-CNN, Cascade R-CNN (first line), YOLOX, DDQ-DETR, and Mask R-CNN (second line) on the RSOD dataset.

2) Results on the UCAS-AOD Dataset: Tables V and VI report the detection results of each method on the UCAS-AOD dataset, further demonstrating the effectiveness of the proposed PKG-Net. For training set ratios of 1/7 and 3/7, our method does not achieve the highest precision in AP_l, and the reasons for this have been previously analyzed. However, PKG-Net leads in other performance metrics. It is important to note that at a training set ratio of 5/7, our method reaches an AP_{50} of 98.5%. It surpasses the AP_{50} scores obtained by other methods on the full training set, confirming that PKG-Net can effectively extract distinctive features of aircraft even with limited sample volumes. Across various training set ratios, compared to DDQ DETR, PKG-Net consistently outperforms AP, with increases of approximately 0.6%, 1.1%, 0.2%, and 1.8%, respectively. Across various training set ratios, compared to RT-DETR, PKG-Net consistently outperforms AP, with increases of approximately 0.6%, 2.9%, 0.7%, and 1.5%, respectively.

We visualize the detection results in Fig. 14. In the first scenario, our method accurately recognizes and precisely delineates the position of each aircraft, with bounding boxes closely matching the ground truth annotations. There are no instances of overlap or omissions, demonstrating high detection precision. In the second scenario, even though the aircraft

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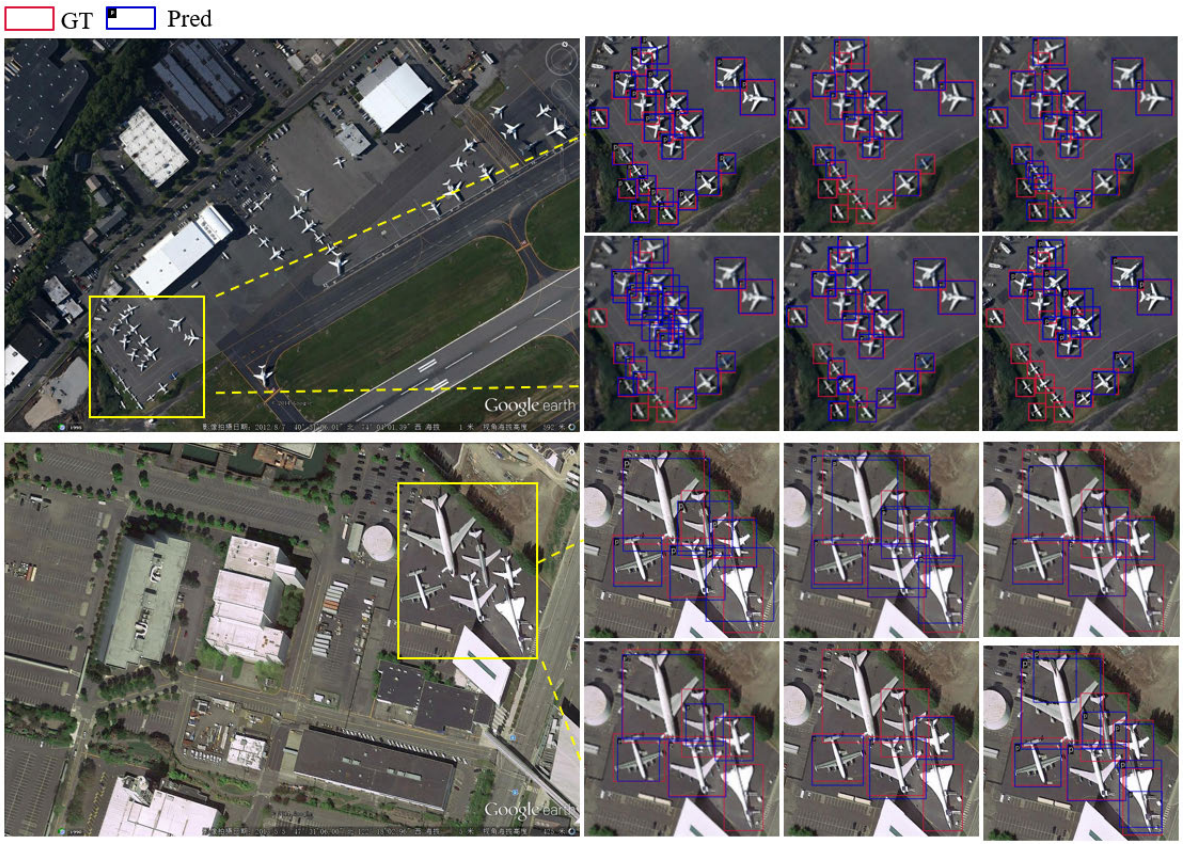
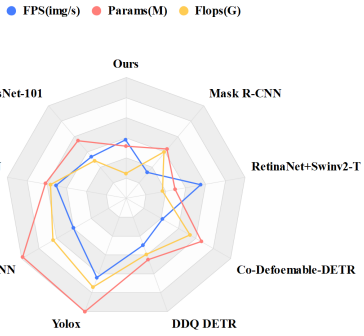


Fig. 14. Partially enlarged comparison of PKG-Net, Faster R-CNN, Cascade R-CNN (first line), YOLOX, DDQ-DETR, and Mask R-CNN (second line) on the UCAS-AOD dataset.



Models	Input Size
RetinaNet+ResNet-101	(800,800)
Faster R-CNN	(1280,800)
Cascade R-CNN	(1280,800)
Yolox	(1280,800)
DDQ DETR	(800,913)
Co-Defoemable-DETR	(1280,800)
RetinaNet+Swinv2-T	(800,800)
Mask R-CNN	(800,928)
Ours	(800,800)

Fig. 15. Running time and model complexity of different methods.

are closely spaced and vary significantly in scale, our method still distinguishes adjacent objects without misidentification. Other methods fall short in the compactness of the bounding boxes and the integrity of object recognition, with some exhibiting noticeable overlaps or missed detections. These visual results adequately demonstrate that, regardless of the scene complexity or object arrangement density, our algorithm consistently delivers high-performance detection.

3) Efficiency Analysis: In the analysis of model efficiency, we consider three key metrics: frames per second (FPS), number of parameters (Params), and floating-point operations per second (Flops). FPS measures the speed at which the model processes images, expressed in images per second (img/s). Params indicate the number of model parameters, measured in millions (M). Flops reflect the number of floating-point operations performed per second, which can be interpreted as computational speed. A low Flops value suggests that the model requires fewer computational resources, making it suitable for devices with limited computing power. From Fig. 15, it is evident that our method demonstrates high efficiency in terms of FPS while maintaining a moderate number of Params and showing lower Flops. This means that our method can achieve fast and energy-efficient detection without sacrificing performance, making it particularly suitable for real-time applications with limited computational resources.

V. Conclusion

This article aims to narrow the gap between the inference logic of end-to-end detectors and the interpretative logic of human experts. Physical knowledge features are initially defined, and their three properties are summarized. Based on this foundation, the CG feature of aircraft is identified, leading to the proposal of the physical knowledge-guided network (PKG-Net). This network utilizes a manually designed MS-CFFM to effectively extract and integrate aircraft CG features from images across various scales. In addition, ACSM is developed for selective activation and weight redistribution among channels, focusing on those that contribute most significantly to learning CG features while maintaining the model's inherent feature extraction capabilities. HAFFM establishes interchannel interaction and coordinates sensitivity within feature maps, emphasizing the intrinsic links and distinctions between different channels to capture a more comprehensive

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array of contextual information. Extensive experiments conducted on aircraft datasets in large and complex scenes show that the model achieves significant performance improvements, even with limited training samples. In comparison to other methods, the model presented in this research stands out for its superior detection accuracy and highly competitive processing speeds. In the future, we will further investigate the combination of physical knowledge features and network design to greatly facilitate the detection of objects in RS images.

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